

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			Any 3 ×(1) from: - The particles get closer together / less room to move around / more densely packed - They collide more frequently - Collide [more frequently] with the walls [of the container] - More collisions mean more force [on less area so more pressure]	3			3		
	(b)	(i)		Any of the following calculations earns the 1st mark: $\frac{20}{-223} = -0.09$ $\frac{40}{-173} = -0.23$ $\frac{80}{-73} = -1.1$ A second calculation plus ‘do not agree’ earns the 2nd mark DO NOT penalise omission of – sign. Award 2 marks for: This is only true if the temperatures are in {kelvin / K}			2	2	2	2
		(ii)		Extend line back (1) Read the T when { $V = 0$ / where line crosses the T axis} (1)	2			2		2
	(c)			$T_2 = 67 + 273 = 340$ (1) Substitution: $\frac{2800}{280} = \frac{V_2}{340}$ ecf (1) Rearrangement: $V_2 = 10 \times 340$ (1) $= 3400 \text{ [cm}^3\text{]}$ (1) Award 3 marks for an answer of 26 800 [cm ³] from use of 67°C	1	1 1 1		4	4	
				Question 6 total	6	3	2	11	6	4